

Amen Woldesenbet Brhanemesqel

SOFTWARE ENGINEER · RESEARCH ASSISTANT

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Education

KAIST (Korea Advanced Institute of Science and Technology)

Daejeon, South Korea

B.S. IN COMPUTER SCIENCE AND BRAIN AND COGNITIVE SCIENCES, WITH A FULL-RIDE SCHOLARSHIP

Aug. 2022 - Present

- **Relevant Coursework:** Introduction to Design, Requirements Engineering for Smart Environments, Biology of Neurons, Neural Circuits for Cognition, Laboratory on Human Brain, Psychological and Behavioral Sciences.

AAiT (Addis Ababa Institute of Technology)

Addis Ababa, Ethiopia

ADDISCODER CODING CAMP (WWW.ADDISCODER.COM)

Jul. 2019 - Aug. 2019

Experience

Interactive Computing Lab (Prof. Uichin Lee), KAIST

Daejeon, South Korea

UNDERGRADUATE RESEARCHER

Jul. 2025 - Current

- Built the Android app system for a Wizard-of-Oz mobile automation agent used in the paper's user study with older adults, with an in-app floating chat overlay and a FastAPI backend with a wizard-side console.
- Quantitatively analyzed smartphone interaction logs (delegation versus intervention, interaction rate, active interaction span).
- Co-authored the IMWUT paper *Speak, Watch, Verify: A Mental Model of Older Adults' First-Use Interactions with Mobile Automation Agents*; under major revision, currently working on the resubmission.
- Prototyped a RAG-based Q&A assistant for older adults on YouTube as a preliminary study that informed the Wizard-of-Oz study.

Make Lab (Prof. Andrea Bianchi), KAIST

Daejeon, South Korea

UNDERGRADUATE RESEARCHER

Jan. 2026 - Current

- Building a Unity-based virtual reality system for spatial audio navigation in situations where vision is unavailable.
- Designing VR study scenarios (path to exit, obstacle search) to test how far audio alone can support navigation, to inform navigation support for visually impaired people.
- Previously built a node-based image generation interface where each visual component is a node with parent-child dependencies on a radial canvas, so edits to a child recursively propagate up the hierarchy.
- The system composes images pairwise through the node graph (two at a time), letting users build scenes from more reference images than the underlying model normally supports while keeping identity drift under control.

Mental Health Accessibility Project, Batoners (CEO Lee Ji-su)

Daejeon, South Korea

UI/UX DESIGN INTERN

Dec. 2024 - Feb. 2025

- Redesigned an existing mental-health app for blind and deaf users, with haptic feedback, text-to-speech output, and visual aids.
- Wrote accessibility documentation for the development team.

Wormery Lab (Prof. Netta Cohen), University of Leeds

Leeds, United Kingdom

RESEARCH INTERN

Jul. 2026 - Aug. 2026

- Studied whether a neural circuit that loses part of its structure to injury can recover its timing using only the connections it still has, motivated by recovery after spinal cord injury.
- Worked in a dynamical-systems framework, characterising limit cycles, phase-response curves and phase locking in pulse-coupled FitzHugh-Nagumo oscillators.
- Found that, within the model, surviving connections alone restored the injured unit's firing rhythm, and discussed the model's limits for recovery after spinal cord injury.

Brain Repair & Organoid Intel Lab (Prof. Minee Choi), KAIST

Daejeon, South Korea

RESEARCH ASSISTANT

Mar. 2024 - Dec. 2024

- Conducted research on induced pluripotent stem cells (iPSCs) and their applications in personalized drug discovery.
- Studied the reprogramming process of somatic cells into iPSCs using Yamanaka factors and analyzed methods for differentiating iPSCs into cortical neurons.

Manuscripts

Speak, Watch, Verify: A Mental Model of Older Adults’ First-Use Interactions with Mobile Automation Agents

IMWUT (PACM on Interactive, Mobile, Wearable and Ubiquitous Technologies)

FOURTH AUTHOR | MAJOR REVISIONS DECISION, REVISION IN PROGRESS

2025 - Current

- Wizard-of-Oz study with 30 older adults (60+) completing 20 everyday smartphone tasks with a voice-driven agent; proposes the Initiation–Intervention–Completion (IIC) model of user breakdowns.

Skills

Programming	Python, Kotlin, JavaScript, C/C++, SQL, HTML/CSS
Frameworks & Tools	Unity, React, Flutter, FastAPI, Docker, Git
ML & Data	RAG pipelines, ChromaDB, vision-language model prompting

Projects

ClothCall: Discreet Clothing Inspection for Blind Adults

Addis Ababa, Ethiopia & Daejeon, South Korea

LEAD DEVELOPER & RESEARCHER

Mar. 2026 - Current

- Ran a formative study with five blind participants in Ethiopia, in Amharic — narrowed scope to stain and fading detection, and surfaced a preference for a trusted person’s judgment over AI analysis alone.
- Led development of the Android app in Kotlin and Jetpack Compose, using a vision-language model.
- Built a trusted-person calibration: a caregiver sets a reference photo and tolerance level, and the app refers to them by name instead of giving a generic AI output.
- Designed a phone-call interaction so the user holds the phone to their ear and hears the result privately.
- Co-authoring a paper on the project’s design contributions.

Modelling Reorganisation and Recovery of Neural Circuits After Injury

University of Leeds, School of Computer Science

RESEARCH INTERN

Jul. 2026 - Aug. 2026

- Built and validated a staged Python model, from a linear control oscillator to a ten-unit neuron chain.
- Chose timesteps by self-convergence and checked the code against the control system’s analytic solution.
- Modelled injury as a cut connection and as a reduced drive current, then compared recovery in each.
- Internship report.

Wizard-of-Oz Mobile Automation Agent for Older Adults

IC Lab, KAIST, Daejeon, South Korea

ANDROID & BACKEND DEVELOPER

Oct. 2025 - Mar. 2026

- Built the Android app system for a Wizard-of-Oz agent: a floating chat overlay that gives older adults voice and text guidance and structured choices on screen.
- Built a wizard console on a FastAPI backend, so a human operator guides users in real time by picking steps and choice templates instead of typing freely.
- Used the system to run the user study underlying the IMWUT manuscript.
- GitHub repo.

RAG-Based Q&A Assistant for Older Adults on YouTube

IC Lab, KAIST, Daejeon, South Korea

DEVELOPER

Jul. 2025 - Aug. 2025

- Prototyped a RAG-based Q&A assistant that gives older adults in-app help on YouTube, combining semantic FAQ retrieval with a small-LLM fallback and a chat overlay built as a Chrome extension.
- Used MiniLM embeddings and ChromaDB over 30 curated FAQs, with FLAN-T5-small as the fallback, served through FastAPI.
- Evaluated response time of FAQ retrieval versus the LLM fallback.
- Demo | GitHub repo.

GrabNGo Mobile App (Introduction to Software Engineering)

Daejeon, South Korea

FRONT-END DEVELOPER

Sep. 2024 - Dec. 2024

- Built the mobile app front end for a campus convenience store checkout system where customers walk in, shop, and walk out without a check-out line.
- Implemented registration, sign-in with password recovery, a welcome screen after face authentication, a live virtual cart, and the payment screen at walkout.
- Paired with the backend team to debug authentication and cart tracking, and tested usability.
- Demo.